THE COMPLEMENT OF A CLOSED POINT IN A SEPARATED IRREDUCIBLE POSITIVE-DIMENSIONAL SCHEME OVER A FIELD IS NOT PROPER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The image of a proper scheme under a separated morphism is closed. Stacks Project, Morphisms, Lemma morphisms-lemma-image-proper-scheme-closed (tag 01W6). Contradicts non-closedness of U to show U is not proper over the base field.

Source reference. Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a separated, irreducible scheme of dimension > 0 over a field k. Let $x \in X$ be a closed point. The open subscheme $X \setminus \{x\}$ is not proper over k.*

Proof. Since X is irreducible, $U = X \setminus \{x\}$ is not closed in X. In particular, the immersion $U \to X$ is not proper. By Morphisms, Lemma 01W6 (here we use X is separated), $U \to \mathrm{Spec}(k)$ is not proper either. □

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